

Simulations & Transmission Efficiencies

GEANT4 UserInput

Beam Parameters:

- 10 # beam Z
- 22 # beam A
- 10 # beam charge state Q
- 66 # beam kinetic energy (MeV)
- 0.2 # fractional energy spread (Sigma %)
- 1.0 # beam-spot diameter (mm)
- 0.2 # normalized transverse emittance (pi mm mrad)

TARGET

IN # IN/OUT
6.0661318 # thickness (um)
0 # target z position offset (cm)
0.8226 # density (g/cm3)
2 # number of elements
3 6.94 0.267 # element 1: Z, M (g/mol), mass fraction
8 18.998 0.7324 # element 2: Z, M (g/mol), mass fraction

Two-body reaction: 1+2->3+4

10 # Z1 beam
22 # A1
3 # Z2 target
7 # A2
12 # Z3 recoil
26 # A3
1 # Z4 ejectile
3 # A4
0.0 # min c.m. angle
180 # max c.m. angle
11 # charge state of recoil
11.32 # excitatβion energy of recoil last: 11.910787

DEGRADER 1 (Carbon Backing)

IN # IN/OUT
0.225 # thickness (um)
2.26 # density (g/cm3)
1 # number of elements
6 12 1 # element 1: Z, M (g/mol), weight ratio

Tracking 26Mg

Commit 881c14f681a4c4a3e2f8b02c081c5b37392fbdab

This is the version of the simulation that outputs 2 files:

Ex_xxMeV_Angleddeg.dat (main data file 13 columns),

SpecEx_xxMeV_angleddeg.dat (Recoil info for transmission efficiency calculation)

Z_{rec}	A_{rec}	E_{lab_rec}	θ_{lab_rec}	Φ_{lab_rec}	θ_{cm_rec}	Z_{eject}	A_{eject}	E_{lab_eject}	θ_{lab_eject}	Φ_{lab_eject}	θ_{cm_eject}	E_{beam}
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Spectrometer angles ϑ, φ :

E_{lab_rec}	ϑ	φ
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$$\vartheta = \arctan(\tan \Theta \cos \Phi)$$

$$\varphi = \arctan(\tan \Theta \sin \Phi)$$

Φ, Θ are spherical coordinate angles

Tracking 3H

Commit 486d8a65c3d3752f5cb7286635d43ecf3d0d6b29

Outputs: S3_triton_ring_observables.dat that contains energies and angles of tritons at:

1. the reaction point
2. at target exit
3. just before hitting the S3
4. Energy of triton with gaussian smear

(see the data file header for detail)

```
// -----
const G4bool kApplyTritonLabAngleGate = true; // Enable reaction-point pre-sampling gate to reduce wasted transport.
const G4bool kUseS3RingGate = false; // Ignored when reaction-point gate is disabled.
const G4int kSelectedS3Ring = 0; // 0-based ring copy number to keep (requested: ring 0).
const G4bool kGenerateTritonToS3 = true; // If true, generate product #4 (triton) as tracked primary.
const G4int kS3RingCount = 24; // Must match S3 geometry ring count.
const G4double kS3InnerRadius = 11.0 * mm; // Must match S3 geometry inner radius.
const G4double kS3OuterRadius = 35.0 * mm; // Must match S3 geometry outer radius.
const G4double kS3DistanceFromTarget = 31. * mm; // Must match upstream S3 target distance.
const G4double kManualGateMinDeg = 130.0; // Manual fallback min theta if ring gate is off.
const G4double kManualGateMaxDeg = 133.0; // Manual fallback max theta if ring gate is off.
const G4int kS3AcceptedRowSamplingBudgetFactor = 200; // Max generated events = factor * requested accepted output rows.
const G4int kReactionBeamOnChunkEvents = 5000; // Reaction is run in chunks; reduces upfront beam pre-generation.
const G4int kBeamTopUpChunkEvents = 5000; // Extra beam samples are generated in small chunks on demand.
// const G4double kManualGateMinDeg = 130.0; // Manual fallback min theta if ring gate is off.
// const G4double kManualGateMaxDeg = 160.0; // Manual fallback max theta if ring gate is off.
```

```
namespace {
// -----
// Hardcoded output gate on triton angle right before S3 entry.
// Set these values directly in code (no UI macro control).
// -----
const G4bool kApplyS3PreHitThetaOutputGate = true; // Enable/disable plain-text row fil
const G4double kS3PreHitThetaMinDeg = 131.531771; // Lower theta bound (deg) for
const G4double kS3PreHitThetaMaxDeg = 132.357455; // Upper theta bound (deg) for
// const G4double kS3PreHitThetaMinDeg = 130.0; // Lower theta bound (deg) for wr
// const G4double kS3PreHitThetaMaxDeg = 160.0; // Upper theta bound (deg) for wr
// const G4double kManualGateMinDeg = 158.83874018; // Manual fallback min theta if rin
// const G4double kManualGateMaxDeg = 160.46334506;

bool IsS3PreHitThetaAccepted(G4double thetaDeg)
{
```

EMMAEventAaction.cc

Two Gates:

EMMAPrimaryGeneratorAction.cc

sets the gate for the sampler,

EMMAEventAaction.cc

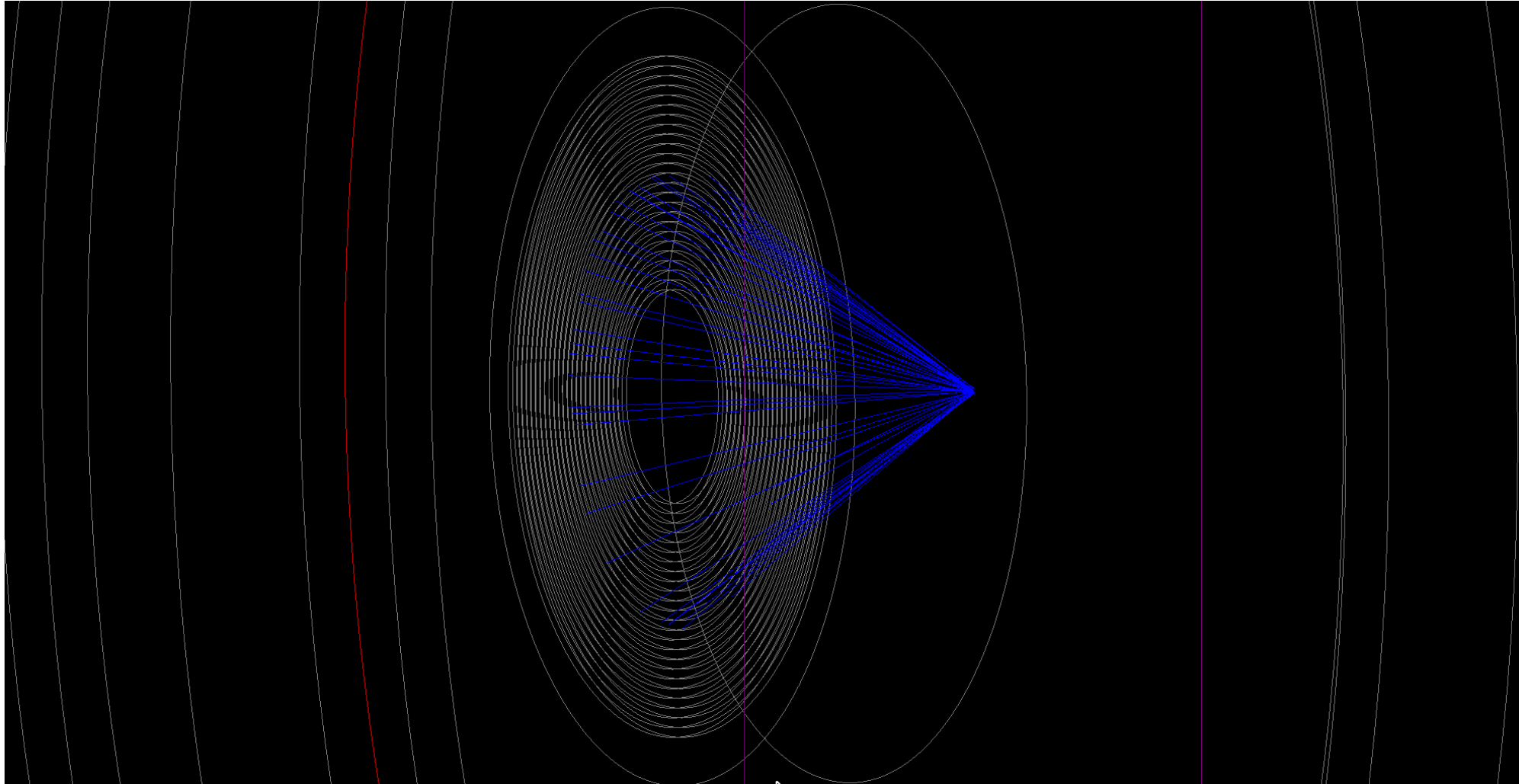
gate selects the tritons that actually hit the rings/within the angle range

EMMAPrimaryGeneratorAction.cc

S3 Construction

24 Rings, [131.5, 160.5] degrees.
Primary tracks: Tritons

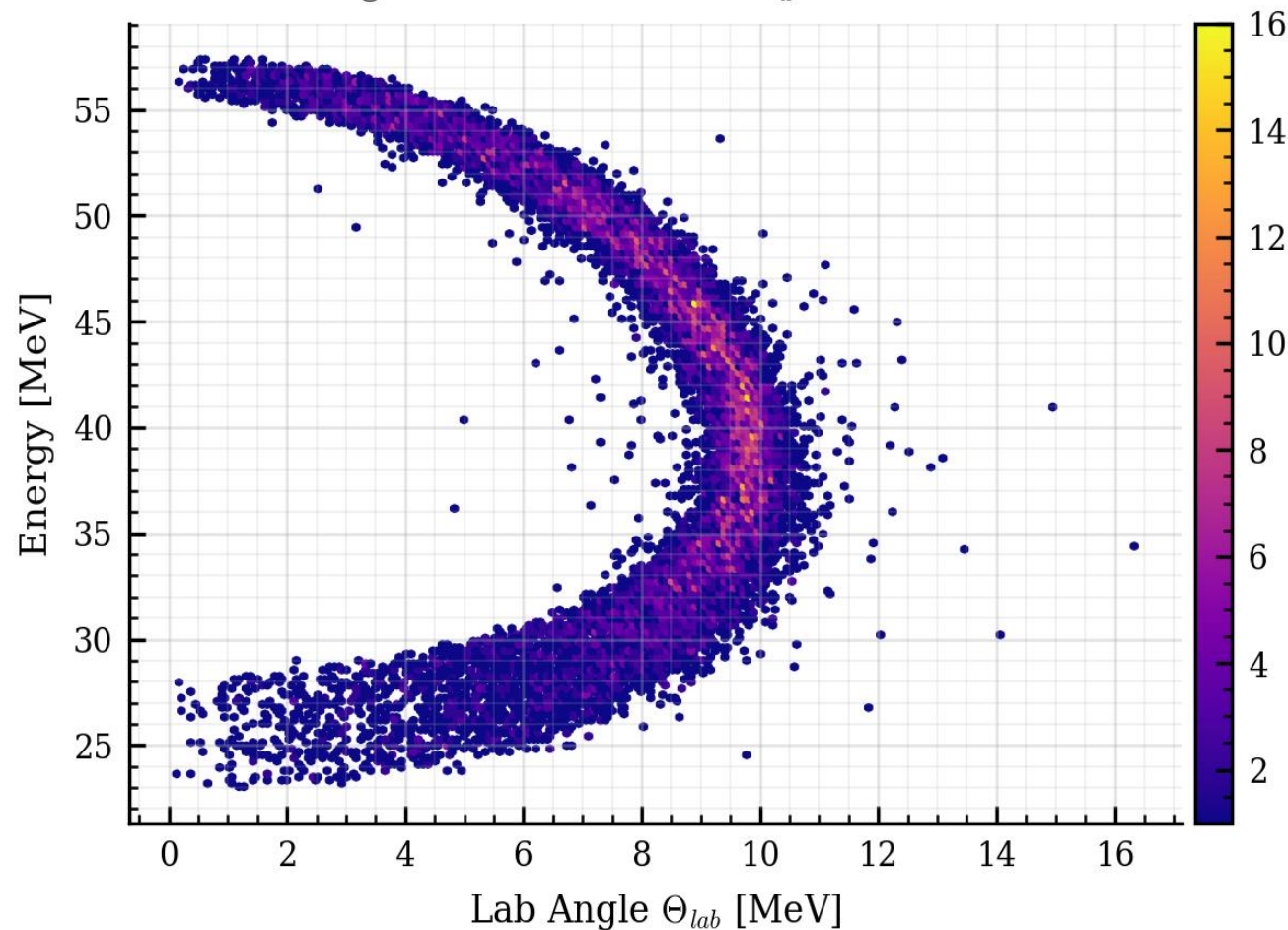
EMMADetectorConstruction.cc, EMMAS3Detector.cc, EMMAS3Hit.hh



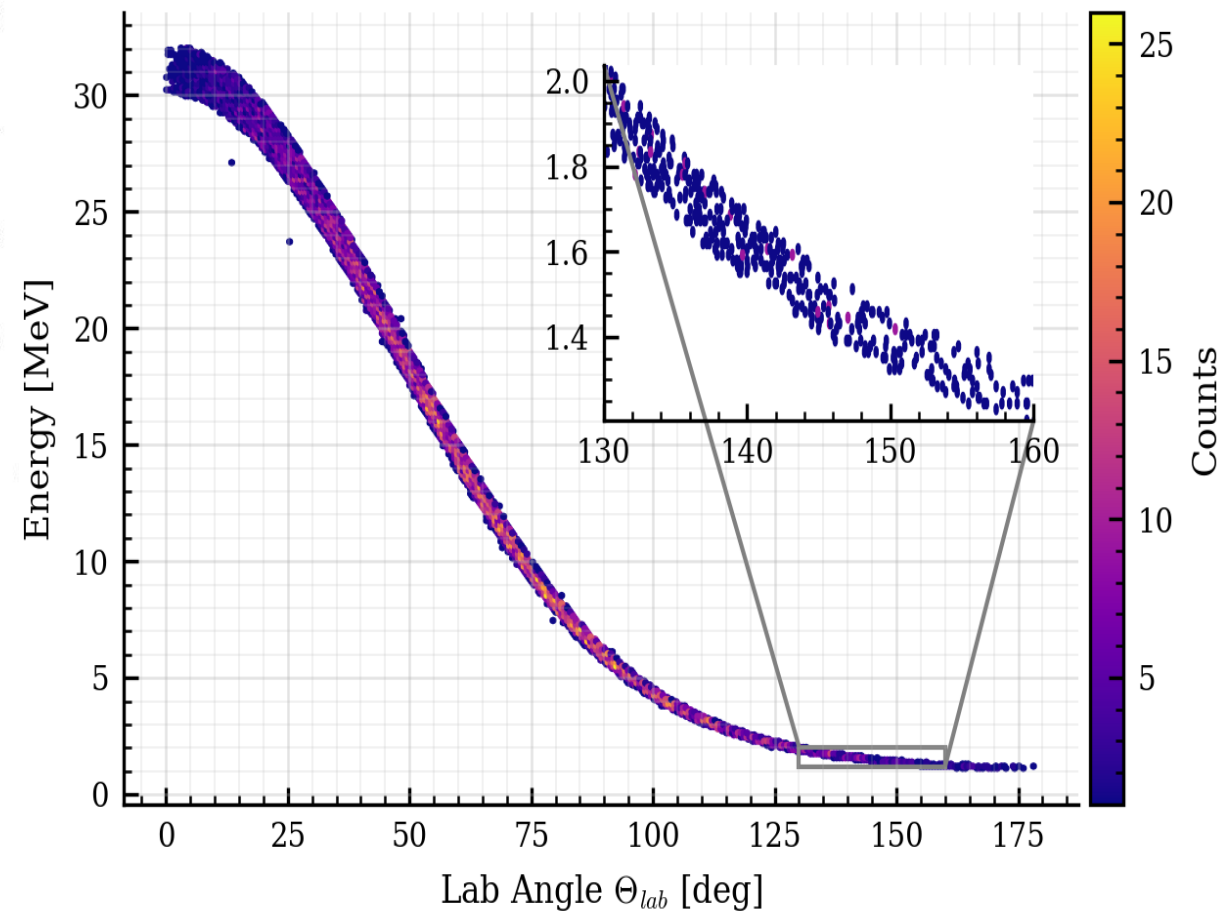
Kinematics

The angles are sampled isotropically

^{26}Mg Kinematics Curve $E_x = 11.32\text{ MeV}$



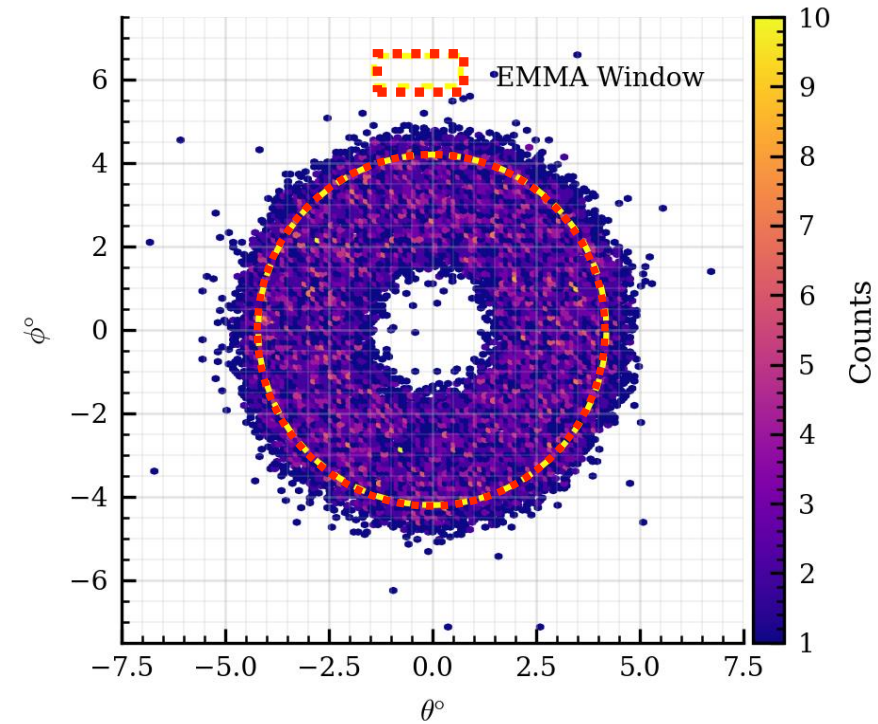
Triton Kinematics Curve $E_x = 11.32\text{ MeV}$



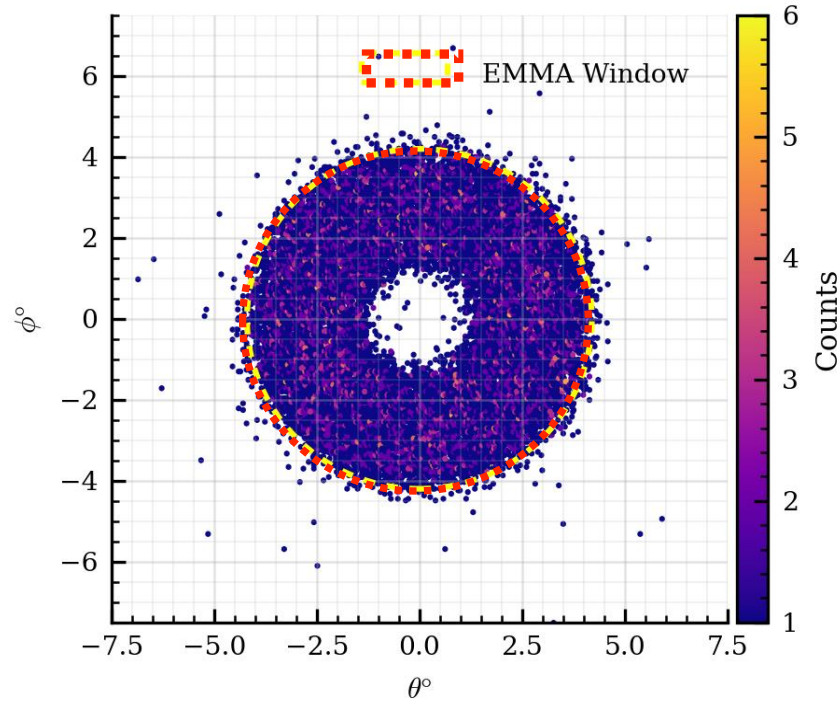
Recoil (^{26}Mg) Distributions

Triton angle $\Theta \in [130^\circ, 160^\circ]$

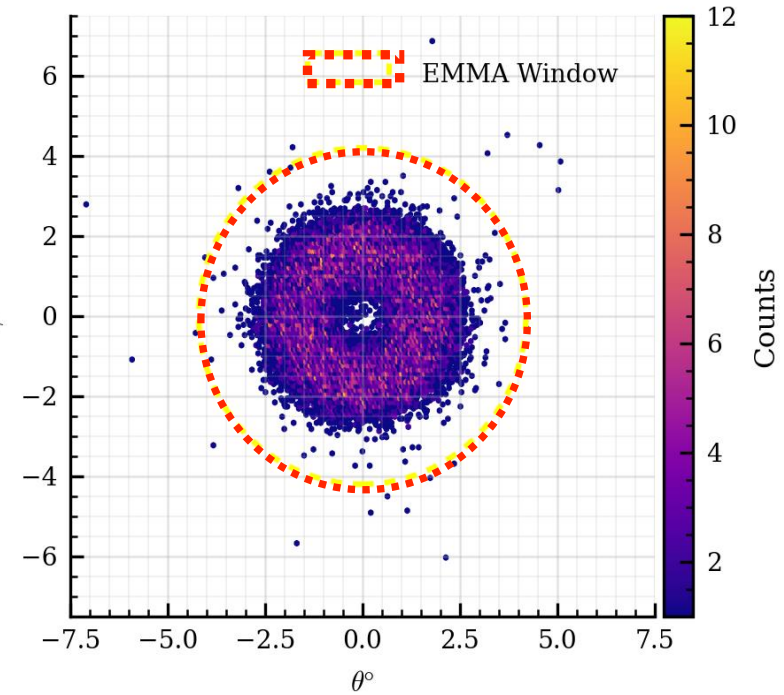
$$E_x = 3.588 \text{ MeV}$$



$$E_x = 6.625 \text{ MeV}$$



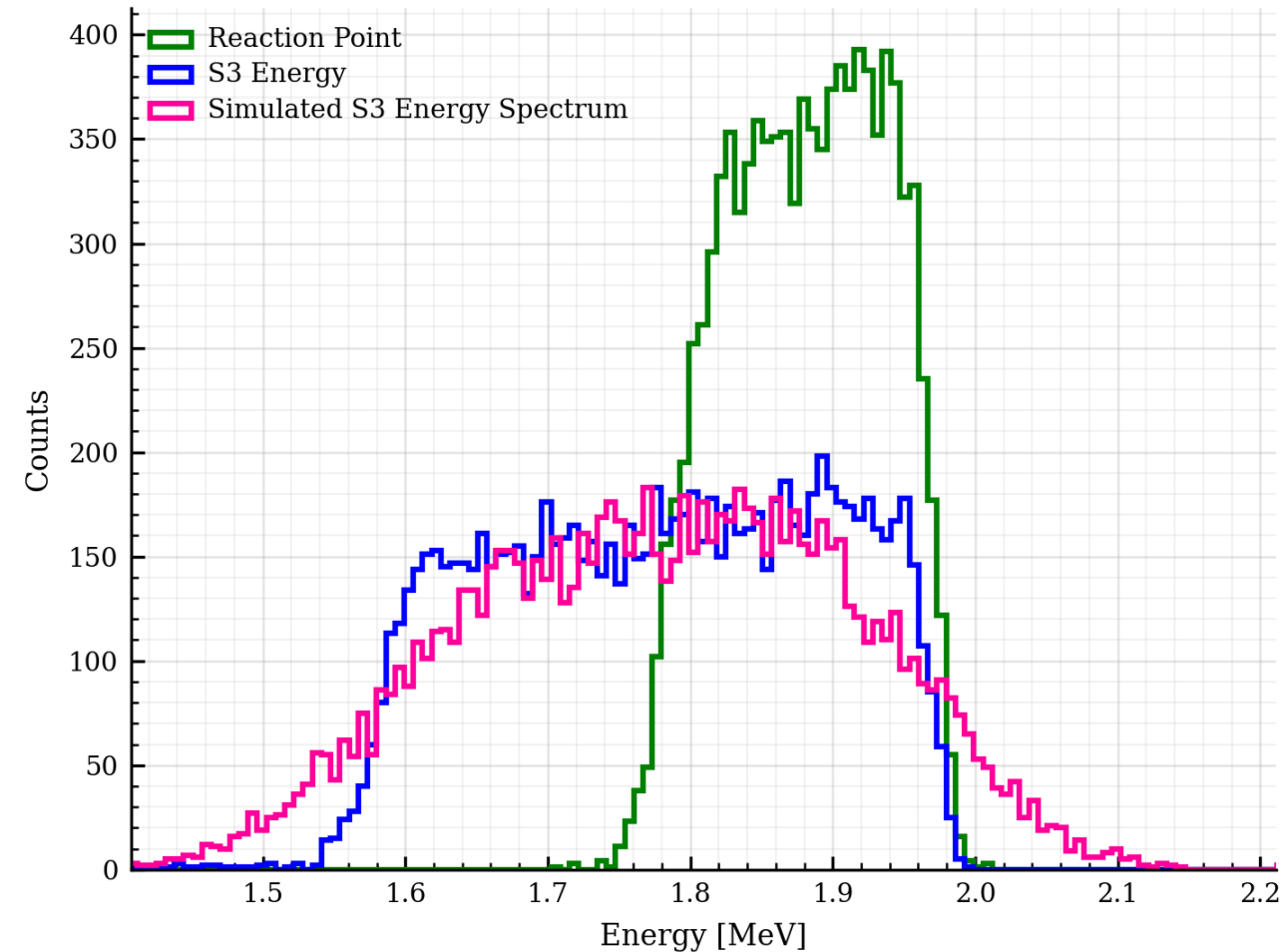
$$E_x = 12.345 \text{ MeV}$$



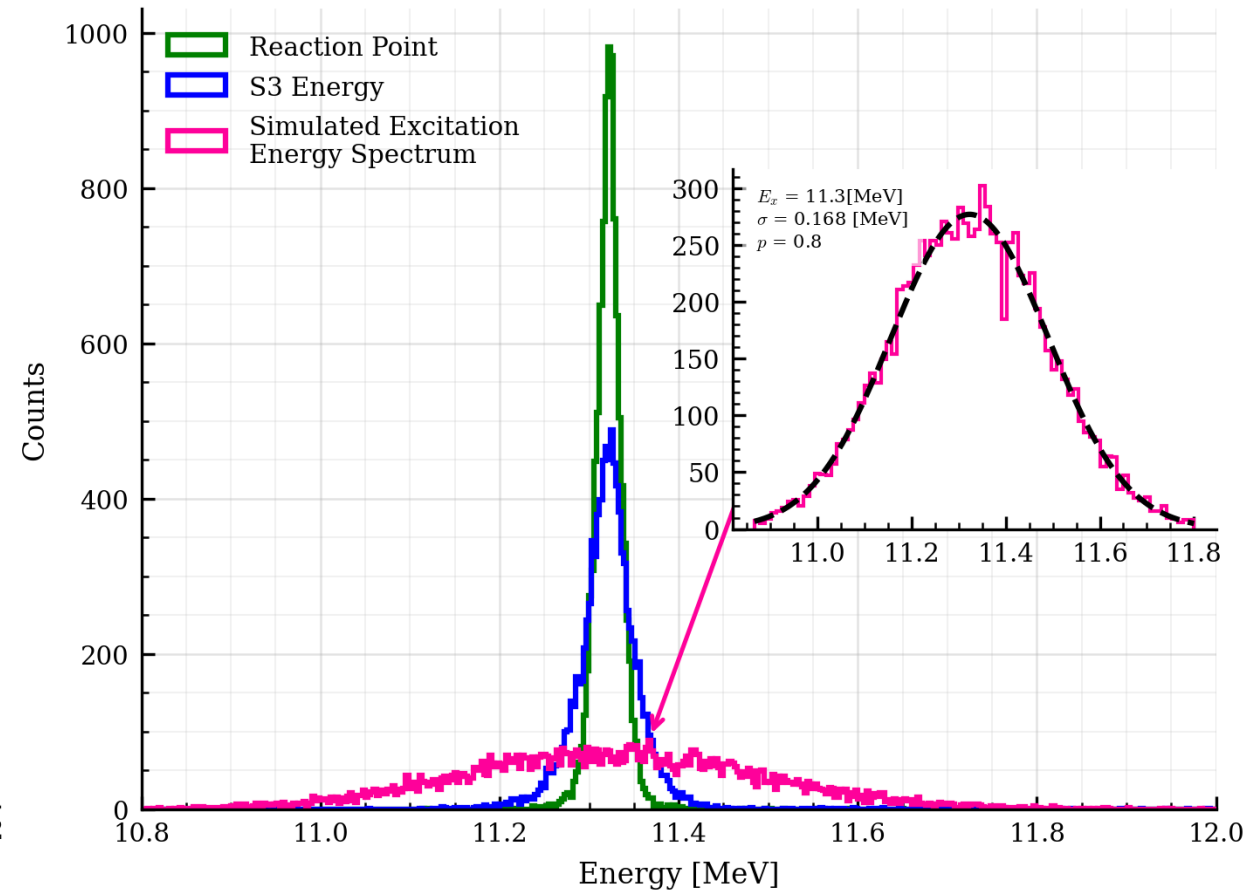
Excitation Energy Reconstruction (11.32 MeV, 1^-)

Here: Gaussian Smear sigma = 70keV

Triton Energy Distribution ($E_x = 11.32$ MeV)



Excitation Energy Distribution Computed from Tritons at Different Points with Energy Loss Corrections ($E_x = 11.32$)



**Need Barry to calculate the Transmission
Efficiencies:**